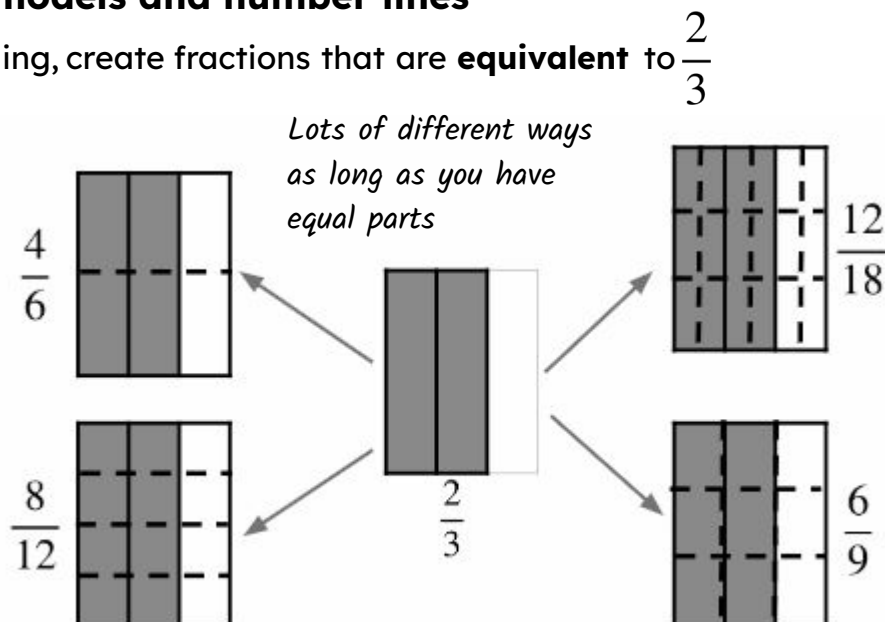




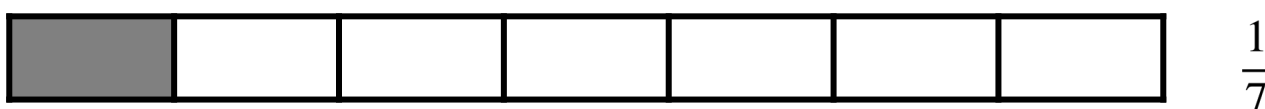
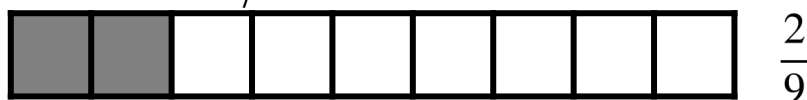
The identity property of multiplication and division

Task A: Bar models and number lines

- 1) By subdividing, create fractions that are **equivalent** to $\frac{2}{3}$



- 2) Sam drew a bar to show $\frac{2}{9}$
 Sam then drew another bar underneath
 which shows $\frac{1}{7}$



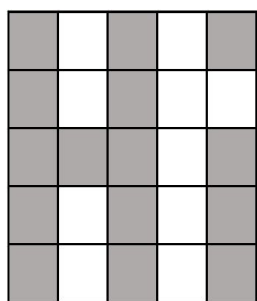
Sam then said, "It shows is **equivalent** $\frac{2}{9}$ to $\frac{1}{7}$ "

Is Sam correct? Explain why.

Incorrect. The bar that represents the whole must be the same length to compare fractions this way.

- 3) Andeep says $\frac{1}{2}$ is shaded. Is he correct?

You can show your working out on the diagram.



Andeep is incorrect. Splitting the shape equally, there are 15 out of 25

15 out of 25 shaded is not equivalent to $\frac{1}{2}$

Task B: Property of multiplication

 1) Fill in the gap to show the fraction, working out and **equivalent fraction**

a) $\frac{4}{9} = \frac{4 \times 2}{9 \times 2} = \frac{8}{18}$

c) $\frac{1}{3} = \frac{1 \times 8}{3 \times 8} = \frac{8}{24}$

b) $\frac{7}{10} = \frac{7 \times 4}{10 \times 4} = \frac{28}{40}$

d) $\frac{12}{11} = \frac{12 \times 3}{11 \times 3} = \frac{36}{33}$

 2) Shade in the equivalent fraction to $\frac{1}{3}$ to reveal a picture.

$\frac{2}{6}$	$\frac{1}{3}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{30}{90}$	$\frac{10}{30}$
$\frac{1}{3}$	$\frac{4}{7}$	$\frac{3}{5}$	$\frac{8}{11}$	$\frac{2}{4}$	$\frac{1}{2}$	$\frac{3}{5}$	$\frac{5}{15}$
$\frac{10}{30}$	$\frac{8}{11}$	$\frac{30}{90}$	$\frac{2}{4}$	$\frac{4}{7}$	$\frac{30}{90}$	$\frac{4}{7}$	$\frac{30}{90}$
$\frac{5}{15}$	$\frac{3}{5}$	$\frac{2}{4}$	$\frac{2}{4}$	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{8}{11}$	$\frac{2}{6}$
$\frac{30}{90}$	$\frac{1}{2}$	$\frac{30}{90}$	$\frac{4}{7}$	$\frac{8}{11}$	$\frac{30}{90}$	$\frac{1}{2}$	$\frac{10}{30}$
$\frac{2}{6}$	$\frac{1}{2}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{1}{3}$	$\frac{80}{110}$	$\frac{5}{15}$
$\frac{5}{15}$	$\frac{8}{11}$	$\frac{8}{11}$	$\frac{8}{11}$	$\frac{3}{5}$	$\frac{8}{11}$	$\frac{2}{4}$	$\frac{2}{6}$
$\frac{30}{90}$	$\frac{10}{30}$	$\frac{1}{3}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{1}{3}$



3) Here is a 5 by 5 multiplication grid.

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

Jacob uses it to help write the **equivalent fractions** to $\frac{2}{5}$

He sees

$$\frac{2}{5} = \frac{4}{10} = \frac{6}{15}$$

a) Using the grid, can you find two more **equivalent fractions** to $\frac{2}{5}$?

b) Using the grid, find **equivalent fractions** to $\frac{1}{4}$

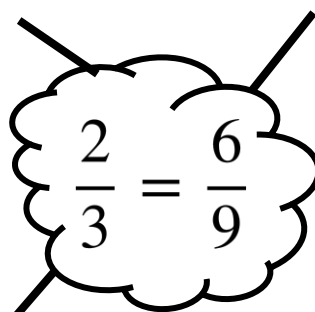
$$\frac{1}{4} = \frac{2}{8} = \frac{3}{12} = \frac{4}{16} = \frac{5}{20}$$

$$\frac{8}{20} \text{ and } \frac{10}{25}$$

4) Show $\frac{2}{3} = \frac{6}{9}$ using numbers lines, bar models and multiplication of 1



$$\frac{2}{3} = \frac{2 \times 3}{3 \times 3} = \frac{6}{9}$$



=

